API Design

Backend: FastAPI app "FoodVerse AI" (version 0.1.0). Interactive docs at http://localhost:8000/docs.

Module	Backend file	Method and path	Input	Output
Home	`main.py`	GET `/`	none	service status
Food recognition	`main.py`	POST `/analyze-food`	multipart form: `file` (image), `description` (optional text)	dish, confidence, serving, calories with range, protein, carbs, fat, health score
Freshness	`freshness.py`	POST `/check-freshness`	multipart form: `file` (image)	freshness result with confidence
Hygiene	`hygiene.py`	GET `/hygiene-checklist`	none	list of checklist items
Hygiene	`hygiene.py`	POST `/hygiene-score`	JSON: `checked` (list of items)	hygiene score
Smart Kitchen	`smart_kitchen.py`	GET `/kitchen/items`	none	list of kitchen items
Smart Kitchen	`smart_kitchen.py`	POST `/kitchen/items`	JSON: `name`, `expiry` (date)	added item
Smart Kitchen	`smart_kitchen.py`	DELETE `/kitchen/items/{item_id}`	path: `item_id` (integer)	deletion result
Recipe	`recipe.py`	POST `/recipe`	JSON: `ingredients`, `dish`	recipe
Voice	`voice.py`	POST `/transcribe`	multipart form: `file` (audio)	transcript
Meal Planner (AI)	`meal_ai.py`	POST `/ai-meal-plan`	JSON: `age`, `gender`, `height_cm`, `weight_kg`, `activity`, `goal`, `diet`, `cuisine`, `allergies`, `conditions`, `history`	AI-generated meal plan (Ollama)
Meal Planner	`meal_planner.py`	POST `/plan-meal` and POST `/meal-plan`	JSON: `goal`, `diet`, `calories` (default 2000)	meal plan

Validation rules (meal planner)

- `age`: 12 to 100 (default 25)
- `height_cm`: 100 to 230 (default 165)
- `weight_kg`: 30 to 250 (default 60)
- Defaults: gender Female, activity Moderate, goal Maintenance, diet Standard, cuisine Global

Invalid input returns HTTP 422 with a validation error.